0.1 Lesbegue Dominated Convergence Theorem

* Riemann-Integral Version

> **Theorem 0.1.1.** Let $g, f_n : (0,\infty) \to \mathbb{R}$, $(n \in \mathbb{N})$ be Riemann-Integrable functions on $[t,T]$ for any $0 < t < T < \infty$. If $|f_n| < g$ and $f_n \to f$ uniformly on every compact subset of $(0,\infty)$, and $\displaystyle\int_0^\infty g(x)dx < \infty$, then
>
> $$\lim_{n\to\infty} \int_0^\infty f_n(x)dx = \int_0^\infty \lim_{n\to\infty} f_n(x)dx$$

Proof. First, observe that for each $n \in \mathbb{N}$,

$$\lim_{t\to 0^+} \lim_{T\to\infty} \int_t^T |f_n(x)|\, dx \le \lim_{t\to 0^+} \lim_{T\to\infty} \int_t^T g(x)\, dx \le \int_0^\infty g(x)\, dx < \infty.$$

Thus, for any $n \in \mathbb{N}$, the improper integral $\displaystyle\int_0^\infty f_n(x)dx$ converges.

Since $f_n \to f$ uniformly on every compact subset $[t,T] \subset (0,\infty)$, f is Riemann-integrable on $[t,T]$ and we have:

$$\lim_{n\to\infty} \int_t^T f_n(x)\, dx = \int_t^T f(x)\, dx.$$

Moreover, the sequence of functions $F_n(t) = \int_t^T f_n(x)\, dx$ converges to $F(t) = \int_t^T f(x)\, dx$ uniformly on $(0,T)$, because

$$\sup_{t\in(0,T)} \left| \int_t^T f_n - \int_t^T f \right| \le \sup_{t\in(0,T)} \int_t^T |f_n - f| \le \sup_{x\in[0,T]} |f_n(x) - f(x)| \cdot T \to 0 \text{ as } n \to \infty.$$

Since 0 is a limit point of $(0,T)$, we can swap the limits as follows:

$$\lim_{t\to 0^+} \lim_{n\to\infty} \int_t^T f_n(x)\, dx = \lim_{n\to\infty} \lim_{t\to 0^+} \int_t^T f_n(x)\, dx.$$

This implies $\int_0^T f(x)\, dx = \lim_{n\to\infty} \int_0^T f_n(x)\, dx$. Similarly, taking $T \to \infty$, we obtain:

$$\int_0^\infty f(x)\, dx = \lim_{T\to\infty} \lim_{n\to\infty} \int_0^T f_n(x)\, dx = \lim_{n\to\infty} \lim_{T\to\infty} \int_0^T f_n(x)\, dx = \lim_{n\to\infty} \int_0^\infty f_n(x)\, dx. \tag{$*$}$$

Rigorous Comment for $(*)$: For each $n \in \mathbb{N}$, $\lim_{T\to\infty} \int_0^T f_n(x)\, dx$ exists; let this be A_n. To show that $\int_0^\infty f(x)\, dx$ exists, we verify the convergence of $\int_0^\infty |f(x)|\, dx$. Suppose $\lim_{T\to\infty} \int_0^T |f(x)|\, dx$ does not exist. Since $\int_0^T |f(x)|\, dx$ is monotonically increasing with respect to T, the only possibility is that it diverges to ∞. Then there exists $R > 0$ such that:

$$\int_0^R |f(x)|\, dx > \int_0^\infty g(x)\, dx.$$

However, since $f_n \to f$ uniformly on $[t,R]$, for any $0 < t < R$, we can choose $N \in \mathbb{N}$ such that for $n \ge N$:

$$\begin{aligned} &\int_0^R |f(x)|\, dx > \int_0^R |f_n(x)|\, dx > \int_0^\infty g(x)\, dx \quad (1) \\ \text{or} \quad &\int_0^R |f_n(x)|\, dx > \int_0^R |f(x)|\, dx > \int_0^\infty g(x)\, dx \quad (2) \end{aligned}$$

In either case (1) or (2), it contradicts the assumption $|f_n| \le g$. Thus $\int_0^\infty |f(x)|\, dx$ converges, and consequently $\int_0^\infty f(x)\, dx$ converges. The rest of the proof follows from Theorem 7.11. $\square$

Lemma 0.1.2. $\mathbb{R}^2$의 회전변환은 선형변환이다.

Proof. 직접 생각해보기.
다시 말해서, 함수 $T : \mathbb{R}^2 \to \mathbb{R}^2$이 임의의 점을 입력받아 각도만 변화시키는 함수라면, 이 함수는 선형사상의 정의를 만족한다. □

Theorem 0.1.3. 행렬 A가 각을 θ만큼 회전시키는 회전변환이라면,

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

이다.

Proof. 점 (x', y')을 점 (x, y)에 대해 θ만큼 각 변환을 한 점이라 하자. 그럼 다음과 같이 쓸 수 있다.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} r\cdot\cos(\alpha+\theta) \\ r\cdot\sin(\alpha+\theta) \end{bmatrix} = r\cdot\begin{bmatrix} \cos(\alpha)\cos(\theta) - \sin(\alpha)\sin(\theta) \\ \cos(\alpha)\sin(\theta) + \sin(\alpha)\cos(\theta) \end{bmatrix} = r\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \cos(\alpha) \\ \sin(\alpha) \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} r\cdot\cos(\alpha) \\ r\cdot\sin(\alpha) \end{bmatrix}$$

여기서 $x = r\cdot\cos(\alpha),\ y = r\cdot\sin(\alpha)$ 이다. 그럼, 마지막 식은 결국

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

이므로, 증명되었다. □